

SOC Internship Day 34 – Detection Engineering

Assessment

Overview

Detection Engineering is the process of creating, tuning, and maintaining security detections that help Security Operations Centers (SOCs) identify malicious activities quickly. Effective detections reduce attacker dwell time and improve incident response efficiency.

Objectives

- Design enterprise-grade security detections.
- Improve SOC visibility against modern threats.
- Reduce attacker dwell time.
- Enhance detection coverage for ransomware, credential theft, and malicious PowerShell activity.

1. Suspicious PowerShell Execution

Purpose

Detect malicious PowerShell commands used for execution, persistence, or downloading payloads.

MITRE ATT&CK Mapping

- T1059.001 – PowerShell

Log Sources

- Windows Event Logs
- PowerShell Operational Logs
- Sysmon

Severity

● High

Detection Indicators

- Encoded PowerShell commands
- DownloadString or Invoke-WebRequest usage
- Hidden execution flags

False Positives

- Legitimate administrative scripts
- IT automation tools

Response Actions

- Isolate affected host
- Investigate command history
- Block malicious scripts

2. Brute Force Authentication Attack

Purpose

Detect repeated failed login attempts targeting user accounts.

MITRE ATT&CK Mapping

- T1110 – Brute Force

Log Sources

- Active Directory Logs
- VPN Logs
- Authentication Servers

Severity

 High

Detection Indicators

- Multiple failed logins from same IP
- Password spraying patterns
- Rapid authentication attempts

False Positives

- User typing incorrect passwords
- Misconfigured applications

Response Actions

- Lock account temporarily
- Investigate source IP
- Enable MFA

3. Privilege Escalation Activity

Purpose

Identify attempts to gain elevated privileges.

MITRE ATT&CK Mapping

- T1068 – Exploitation for Privilege Escalation
- T1134 – Access Token Manipulation

Log Sources

- Windows Security Logs
- Sysmon
- EDR Solutions

Severity

● Critical

Detection Indicators

- Unexpected admin account creation
- Privilege assignments
- Token manipulation events

False Positives

- Approved administrative changes
- System maintenance activities

Response Actions

- Validate privilege changes
- Disable suspicious accounts
- Conduct forensic analysis

4. Data Exfiltration Attempt

Purpose

Detect unauthorized transfer of sensitive data.

MITRE ATT&CK Mapping

- T1041 – Exfiltration Over C2 Channel
- T1020 – Automated Exfiltration

Log Sources

- Firewall Logs

- Proxy Logs
- DLP Solutions
- Network Monitoring Tools

Severity

 Critical

Detection Indicators

- Large outbound data transfers
- Connections to unknown destinations
- Abnormal upload activity

False Positives

- Legitimate backups
- Cloud synchronization services

Response Actions

- Block suspicious connections
- Investigate transferred files
- Notify management

5. Ransomware Behavior Detection

Purpose

Detect behaviors commonly associated with ransomware attacks.

MITRE ATT&CK Mapping

- T1486 – Data Encrypted for Impact

Log Sources

- EDR Logs
- File Monitoring Solutions
- Windows Event Logs

Severity

 Critical

Detection Indicators

- Mass file encryption
- Shadow copy deletion
- Rapid file modifications

False Positives

- Bulk file operations
- Backup software activities

Response Actions

- Isolate infected endpoints
- Stop malicious processes
- Initiate incident response procedures

Detection Workflow

Log Collection



Event Normalization



Detection Rule Execution



Alert Generation



SOC Analyst Review



Investigation & Validation



Containment / Remediation



Post-Incident Review

Tuning Recommendations

- Baseline normal user and system behavior.
- Whitelist trusted administrative tools.
- Implement threshold-based detections.
- Correlate multiple events before alerting.
- Continuously review and tune detection rules.
- Integrate threat intelligence feeds.
- Regularly validate detections through testing.

Executive Summary

This Detection Engineering Assessment focuses on strengthening SOC capabilities against ransomware, credential theft, privilege escalation, malicious PowerShell execution, and data exfiltration. By implementing well-tuned detections aligned with the MITRE ATT&CK framework, organizations can significantly improve threat visibility, reduce false positives, and accelerate incident response times. Proper monitoring, alert triage, and continuous tuning are essential for maintaining an effective security posture.